

## Yuqi Tan

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### EDUCATION

|                    |                                                                                                                                                 |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Apr 2026 - now     | <b>Stanford University, USA</b><br>Instructor                                                                                                   |
| Apr 2021-Mar 2026  | <b>Stanford University, USA</b><br>Postdoctoral Fellow with Dr. Garry Nolan                                                                     |
| Aug 2015-May 2021  | <b>Johns Hopkins University, USA</b><br>PhD with Dr. Patrick Cahan                                                                              |
| Sept 2010-Jul 2014 | <b>The Chinese University of Hong Kong, Hong Kong SAR</b><br>Bachelor of Science in Cell Molecular Biology, First Class Honors<br>Minor: French |

### PERSONAL STATEMENT

I am a computational biologist specializing in spatial multi-omics, integrating expertise in machine learning with a deep understanding of molecular and cellular biology. My work focuses on developing scalable machine learning models and streamlined computational frameworks to tackle complex biological questions, such as improving stem cell engineering, identifying novel cell types, finding spatial biomarkers for cancer immunotherapy response, and discovering spatial mechanisms in neurological disease. I have led and will continue to lead a research team to develop streamlined, scalable machine learning models and apply them to explore the spatial organization of cell types in healthy aging and diseases, leveraging spatial multi-omics and 3D pathology. Through directly mentoring nine students, designing/teaching three custom courses, and contributing to three global health projects, I remain committed empowers future researchers.

### PUBLICATIONS

**First/second author (\* indicates co-first authorship, \$ indicates corresponding author)**

1. **Tan\***✉, Kempchen\*, Becker\*, Haist, Feyaerts, Xiao, Su, Rech, Hickey, and Nolan✉. SPACEc: A Streamlined, Interactive Python Workflow for Multiplexed Image Processing and Analysis. *Nat Comm*. 2025 Nov 27;16(1):10652. PMID: 41309581
2. **Tan** and Cahan. SingleCellNet: a computational tool to classify single cell RNA-Seq data across platforms and across species. *Cell Syst*. 2019 Aug 28;9(2):207-213.e2. PMID: 31377170
3. **Tan\***, Kim\*, Mujal, Chen, Weis, Bergaggio, Micevic, Xie, Park, Hor, Papanicolaou, Shobaki, Pomizi, Delconte, Vendramin, Hedge, Han, Su, Hacohen. Training Tomorrow's Leaders in Cancer Immunology. *Cancer Immunology Res*. 2026 Feb 3;14(2):186-193. PMID: 41524276
4. Brbic\*, Cao\*, Hickey\*, **Tan**, Snyder, Nolan†, and Leskovec†. Annotation of Spatially Resolved Single-cell Data with STELLAR. *Nat Methods*. 2022 Nov;19(11):1411-1418. PMID: 36280720
5. Hickey, **Tan**, Goltsev, and Nolan. Strategies for Accurate Cell Type Identification of CODEX Multiplexed Imaging Data. *Front. Immunol*. 2021 Aug 13;12:727626. PMID: 34484237
6. Kumar, **Tan**, and Cahan. Understanding development and stem cells using single cell-based analyses of gene expression. *Development*. 2017 Jan 1;144(1):17-32. PMID: 28049689

#### BioRxiv/Under Review

1. Azher, Nolan†, **Tan\***. Target Selection Beats Model Size in Virtual Spatial Transcriptomics. *Under Review in JTC*. Jun 2026.
2. **Tan\***, Golstev, Aimnu, Rao, Nolan†. Academic and Commercial Agentic AI Landscape for Spatial Omics Discovery. *Under Review in JTC*. Jun 2026.
3. Zhang\*, **Tan\***, Synder, Nolan† and Ma†. Fault-tolerant 3D reconstruction from 2D spatial proteomics sections. *BioRxiv. Submitted*. Jun 2026.
4. Haist\*, Matusiak\*, **Tan\***, Zimmer, Stege, Kempchen, Mischke, Chu, Weidentahler-Barth, Barlow, Rogall, Gonzalez, Vaertsch, Goltsev, Grabbe, Hickey, and Nolan. Spatially organized inflammatory myeloid-CD8+ T cell aggregates linked to Merkel-cell Polyomavirus driven Reorganization of the Tumor Microenvironment. *BioRxiv. Under Review in Nat Comm*. Feb 2026.
5. Xiao\*, Su\*, **Tan\***, Deng, Li, Ma, Kim, Bai, Liu, Rosoklija, Sissoko, Wu, Santiago, Dwork, Hen, Mann, Fan†, Leong†, Boldrini†. Spatially resolved multi-omic profiling of human hippocampus reveals

region-specific alterations in major depressive disorder. *Under Review* in **Nat. Neuroscience**. May 2025.

6. Tan, Spangler, Han, Farid, Peng, and Cahan. Quantitative comparison of in vitro and in vivo embryogenesis at a single cell resolution. *BioRxiv*. *Under Review* in **Development**. Feb 2024.
7. Kharbanda\*, Tubelleza\*, Tan, Tan, Janke, Sebina, Belz, Kulasinghe, Salim. bgnorm: A Generative Statistical Framework for Background Correction, Normalisation, and Quality Control in Multiplex Spatial Proteomics. *Under Review* in **Nat Method**. Jul 2026

## Co-author

1. Tao, Gong, Tan, Zhu, Snyder, Yu. Diffeomorphic Deformation Refinement via Cell Correspondence Learning for Cross-Modal Whole Slide Image Registration. *Under Review* in **AAAI 2027 Conference**
2. Han\*, Zhu\*, Ruan\*, Zhao, Tan, Monte, Wei, Karathanos, Laquindanum, Charville, Lin, Ling, Ford, Bi, Nolan, Wang. Space-map: 3D single-cell atlas reconstruction from serial sections. *Under Review* in **Nat Method**. May 2026
3. Shaban\*, Chang\*, Qiu, Yeo, Song, ..., Tan..., Ma, Jiang, Mahmoon. A Foundation Model for Spatial Proteomics. *Under Review* in **Nature**. arXiv:2506.03373. Jun 2026.
4. Tan, Yin, Tan, Ivanov, Plucinska, Ilanges, Herb, Wang, Kosse, Cohen, Lin, Friedman. Leptin Activated Hypothalamic BNC2 Neurons Acutely Suppress Food Intake. **Nature**. 2024 Dec;636(8041):198-205. PMID: 39478220
5. Phongpreecha, Ghanem, Reiss, Oskotsky, Mataraso, De Francesco, Reincke, Espinosa, Ng, Costello, Sequoia, Razdan, Xie, Berson, Kim, Seong, Szeto, Myers, Gu, Feister, Verscaj, Rose, Sin, Oskotsky, Roger, Shu, Shome, Yang, Tan, Levitte, Wong, Gaudilliere, Angst, Montine, Kerner, Keller, Shaw, Sylvester, Fuerch, Chock, Gaskari, Stevenson, Sirota, Prince, Aghaeepour. AI-guided precision parenteral nutrition for neonatal intensive care units. **Nat Medicine**. 2025 Jun;31(6):1882-1894. PMID: 40133525
6. Tal, Gunawardana-Zeigler, Peng, Tan, Munoz-Perez, Offenbacher, Kastner, Ciero, Randolph, Gong, Deng, Cahan, Loeb. Inhibition of DKK-1 by WAY262611 Inhibits Osteosarcoma Metastasis. **Mol Cancer Ther**. 2025 May 2;24(5):728-739. PMID: 39781890
7. Offenbacher, Jackson, Hayashi, Zhang, Peng, Tan, Stewart, Ciero, Foley, Casero, Cahan and Loeb. Polyamine Depletion by D, L-alpha-difluoromethylornithine Inhibits Ewing Sarcoma Metastasis by Inducing Ferroptosis. **Clinical Cancer Res**. 2025 Oct 1;31(19):4196-4210. PMID: 40705089
8. Mataraso, Espinosa, Seong, Reincke, Berson, Reiss, Kim, Ghanem, Shu, James, Tan, Shome, Stelzer, Feyaerts, Wong, Shaw, Angst, Gaudilliere, Stevenson. A machine learning approach to leveraging electronic health records for enhanced omics analysis. **Nat Mach Intell**. 2025;7(2):293-306. PMID: 40008295
9. Mirzaei, Hiller, Stelzer, Thiele, Tan, Becker. Computational Approaches for Connecting Maternal Stress to Preterm Birth. **Clin Perinatol**. 2024 Jun;51(2):345-360. PMID: 38705645
10. Yau, Hung, Wong, Huang, Wong, Chow, Tsoi, Tan, Li, Wing, Lai. INSIGHT: An accessible multi-scale, multi-modal 3D spatial biology platform. **Nat Comm**. 2024 Dec 30;15(1):10888. PMID: 39738072
11. Hickey, Haist, Horowitz, Caraccio, Tan, Rech, Baertsch, Rovira Clave, Zhu, Vasquez, Barlow, Agmon, Goltsev, Sunwoo, Covert, and Nolan. T Cell Mediated Curation and Restructuring of Tumor Tissue Coordinates an Effective Immune Response. **Cell Rep**. 2023 Dec 26;42(12):113494. PMID: 38085642
12. Haist\*, Stege\*, Rogall, Tan, Wasielewski, Klespe, Meier, Mohr, Kähler, Wichenthal, Schadenfdorf, Ugurel, Lodde, Zimmer, ..., Grabbe and Loquai. Treatment management for BRAF-mutant melanoma patients with tumor recurrence upon adjuvant therapy: A multicenter study from the prospective skin cancer registry ADOREG. **J Immunother Cancer**. 2023 Sep;11(9):e007630. PMID: 37730278
13. Becker, Nassar, Espinosa, Stelzer, Fayaerts, Berson, Bidoki, Chang, Saarunya, Culos, De Francesco, Fallahzadeh, Liu, Kim, Marić, Mataraso, Payrovnaziri, Phongpreecha, Ravindra, Stanley, Shome, Tan, Thuraiappah, Xenochristou, Xue, Shaw, Stevenson, Angst, Gaudilliere, Aghaeepour. Large scale correlation network construction for unraveling the coordination of complex biological systems. **Nat Comp. Sci**. 2023 Apr;3(4):346-359. PMID: 38116462
14. Strasser, Gibbs, Gascard, Bons, Hickey, Schürch, Tan, Black, Chu, Ozkan, Basisty, Sangwan, Rose, Shah, Camilleri-Broet, Fiset, Bertos, Berube, Djambazian, Li, Oikonomopoulos, Rachael Fels-Elliott, Vernovsky, Shimshoni, Collyar, Russell, Ragoussis, Stachler, Goldenring, McDonald, Ingber, Schilling, Nolan, Tlsty, Huang, and Ferri. Concerted epithelial and stromal changes during

- progression of Barrett's Esophagus to invasive adenocarcinoma exposed by multi-scale, multi-omics analysis. *Dev Cell*. 2025 Oct 20;60(20):2807-2824.e7. PMID: 40695287
15. Becker, Fayaerts, Stelzer, Berson, Chang, Clark, Marić, Culos, De Francesco, Espinosa, Kim, Maric, Mataraso, Payrovnaziri, Phongpreecha, Ravindra, Stanley, Shome, **Tan**, Thuraiappah, Xue, Shaw, Stevenson, Angst, Gaudilliere, Aghaeepour. CoDi: Profiling functional disruption in biological systems. *Under Review in Bioinformatics*. 2022.
  16. Liu, Li, Zhang, van Sambeek, Appleton, Chiang, Ng, Khoshakhlagh, Chen, Garcia-Corral, Wu, Huang, **Tan**, Chao, Aach, Tam, Lim, Raychaudhuri, and Church. Iterative transcription factor screening enables rapid generation of microglia-like cells from human iPSC. *Nat Comm*. 2025 Jun 10;16(1):5136. PMID: 40494892
  17. Becker, Dai, Chang, Fayaerts, Stelzer, Zhang, Berson, Clark, De Francesco, Espinosa, Kim, Marić, Mataraso, Payrovnaziri, Phongpreecha, Ravindra, Shome, **Tan**, Thuraiappah, Xue, Mayo, Quaintance, Laborde, King, Gotlib, Wong, Angst, Shaw, Stevenson, Gaudilliere, Aghaeepour. Revealing the impact of lifestyle stressors on the risk of adverse pregnancy outcomes with multitask machine learning. *Front Pediatr*. 2022 Dec 13;10:933266. PMID: 36582513
  18. Bian, Cheng, Su, **Tan**, Kim, Wang, Yoo, Blackshaw and Cahan. Signaling and transcriptional networks governing late synovial joint development. *Dev Biol*. 2026 Jun 4:537:28-38. PMID: 42248342
  19. Velazquez, LeGraw, Moghadam, **Tan**, Kilbourne, Hislop, Liu, Cats, de Sousa Lopes, Plaisier, Cahan, Kiani, and Ebrahimkhani. Gene Regulatory Network Analysis and Engineering Directs Development and Vascularization of Multilineage Human Liver Organoids. *Cell Syst*. 2020 Nov 30; S2405-4712(20)30420-8. PMID: 33290741
  20. Chen, de Rocha, Cahan, Kubaczka, Hunter, Sousa, Mullin, Fujiwara, Nguyen, **Tan**, Landry, Han, Yang, Lu, Jha, Vo, Zhou, North, and Schlaeger. Transcriptome Dynamics of Hematopoietic Stem Cell Formation Revealed Using a Combinatorial Runx1 and Ly6a Reporter System. *Stem Cell Reports*. 2020 May 12;14(5):956-971. PMCID: PMC7220988
  21. Liu, Cheng, **Tan**, Dong, and Bian. Ligustrazine Prevents Intervertebral Disc Degeneration via Suppression of Aberrant TGFβ Activation in Nucleus Pulposus Cells. *Biomed Res Int*. 2019 Dec 2; 2019:5601734. PMCID: PMC6914881
  22. Radley, Schwab, **Tan**, Kim, Lo, and Cahan. Assessment of engineered cells using CellNet and RNA-Seq. *Nat Protoc*. 2017 12:1089-1102 PMCID: PMC5765439
  23. Mayer, Hu, Wang, Cardenas, **Tan**, Pan, and Bednarek, SCD1 and SCD2 Form a Complex that Functions with the Exocyst and RabE1 in Exocytosis and Cytokinesis. *Plant Cell*. 2017 Oct;29(10):2610-2625. PMCID: PMC5774579
  24. Cui, Zhao, Xie, Wong, Wang, Gao, Ding, **Tan**, Ueda, Zhang, and Jiang. MON1/CCZ1-mediated Rab7 activation regulates tapetal PCD and pollen development in Arabidopsis. *Plant Physiol*. 2017 Jan;173(1):206-218 PMCID: PMC5210713
  25. James, Gu, Ramirez-Vizcarrondo, Hasan, Truszkowski, **Tan**, Oupravanh, Khakhalin, and Aizenman. Valproate-induced neurodevelopmental deficits in *Xenopus laevis* tadpoles. *J Neurosci*. 2015 Feb 18;35(7):3218-29. PMCID: PMC4331635

## FUNDING | SCHOLARSHIP | AWARDS

|      |                                                                        |
|------|------------------------------------------------------------------------|
| 2026 | France-Stanford Center Visiting Junior Scholar Fellowship              |
| 2025 | NIH-Institute of Aging K99/R00 Pathway to Independence Award (Pending) |
|      | MD Anderson – the Next Generation Award                                |
|      | Rising Star in Engineering Health Award                                |
| 2024 | Stanford Postdoctoral BioX Travel Award                                |
| 2023 | Stanford Cancer Institute Women's Cancer Center Innovation Award       |
|      | Life Science Alliance Exchange Grant                                   |
|      | Stanford Postdoctoral Travel Award                                     |
| 2022 | Stanford University Bio-X Travel Award                                 |
|      | Stanford University School of Medicine Dean's Fellowship               |
| 2021 | Deep Learning for Microscopy Image Analysis Workshop Scholarship       |
| 2019 | Johns Hopkins Graduate Student Travel Award                            |
| 2014 | S.H. Ho College Eminent Scholarship 2013/14                            |
|      | Science Faculty Dean's List                                            |
|      | Hong Kong Special Administrative Region Government Reaching Out Award  |

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|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2013 | S.H. Ho College Whole Person Development Gold Medal 13/14<br>Finalist for Rhodes Scholarship (Hong Kong) 2014<br>S.H. Ho College Outstanding Student Scholarship<br>S.H. Ho College Scholarship for Overseas Academic Activity<br>Dedicated Research Exchange and Mentorship (DREAM) Program Fellowship |
| 2012 | Science Faculty Dean's List<br>S.H. Ho College Outstanding Student Scholarship<br>S.H. Ho College Annual Academic Scholarship<br>Yasumoto International Exchange Scholarship<br>Young Chi Wan Exchange Scholarship                                                                                      |
| 2011 | S.H. Ho College Outstanding Service Scholarship<br>S.H. Ho College Outstanding Student Scholarship                                                                                                                                                                                                      |
| 2010 | S.H. Ho College Matriculation Scholarships for Academic Excellence (Mainland)                                                                                                                                                                                                                           |

## CONFERENCE PRESENTATION

### Oral presentation

#### Invited

- **Tan**, NIH Spatial Biology- The Next Frontier Symposium (2026, Rockville, Maryland)
- **Tan**, Purdue University (2026, West Lafayette, Indiana)
- **Tan**, Vanderbilt University (2026, Nashville, Tennessee)
- **Tan**, St Jude's Hospital (2026, Memphis, Tennessee)
- **Tan**, University of California, Irvine (2026, Irvine, California)
- **Tan**, University of Michigan, Ann Harbor (2026, Ann Harbor, Michigan)
- **Tan**, Dartmouth College (2026, Hanover, New Hampshire)
- **Tan**, MD-Anderson Cancer Research Center – The Allison Institute Seminar, (2025, Huston, Texas)
- **Tan**, Helmholtz Munich AI for Health Center, (2025, Munich, Germany)
- **Tan**, University of Basel (2025, Basel, Switzerland)
- **Tan**, University of Würzburg, Institute for System Immunology (2025, Würzburg, Germany)
- **Tan**, The German Cancer Research Center (Heidelberg) (2025, Heidelberg, Germany)
- **Tan**, University of Bonn, Cluster of Excellence ImmunoSensation Symposium (2025, Bonn, Germany)
- **Tan**, MD Anderson Data Science Hackathon (2025, Huston, Texas): TLS & Where to find them
- **Tan**, Visualization and User Experience Seminars on Spatial Biology at the Harvard Medical School – the Future of 3D Tissue Maps (2025, Boston, MA): On the way to a human 3D intestine atlas
- **Tan**, MultiOmics 2024 (2024, Brisbane, Australia): *Make Computational Analysis for Multiplexed Image Easy.*
- **Tan**, EPFL (2024, EPFL, Switzerland): *Versatile and Interactive Python Workflow for 2D and 3D Multiplexed Image Analysis.*
- **Tan**, The European Molecular Biology Laboratory (2024, Heidelberg, Germany): *Best Practice for Multiplexed Image Analysis.*
- **Tan**, Yale University, Biomedical Engineering Department (2023, New Haven, CT): *A Spatial Biologist's Guide to Decode Multiplexed Image Analysis.*
- **Tan**, Columbia University, Biomedical Engineering Department (2023, New York City, NY): *A Spatial Biologist's Guide to Decode Multiplexed Image Analysis.*
- **Tan**, Cancer Grand Challenges Future Leaders Conference 2022 (2022, Barcelona, Spain): *Universal Cancer Knowledge Encyclopedia*
- **Tan**, East China Normal University (2021, Shanghai, China): *Machine-learning tools on multi-omics analysis.*
- **Tan**, Johns Hopkins University Biochemistry Cellular Molecular Biology Program Annual Retreat (2019, Cambridge, MD): *SingleCellNet: a computational tool to classify single cell RNA-Seq data across platforms and across species.*
- **Tan**, Invited talk at Cell Molecular Biology Symposium at Chinese University of Hong Kong (2018, Hong Kong, China): *A quantitative solution of cell type identity for stem cell engineering.*



Selected

- **Tan**, The European Society for Spatial Biology Annual Meeting (2025, Heidelberg, Germany): AI and spatial-omics analysis
- **Tan**, Single Cell Genomics Gordon Research Seminar (2024, Les Diablerets, Switzerland): *Versatile and Interactive Python Workflow for 2D and 3D Multiplexed Image Analysis*.
- **Tan**, Hong Kong Laureate Forum (2023, Hong Kong, China): *Identify structural and cellular drivers of DCIS progression*.
- **Tan**, Artificial Intelligence in Medicine (2022, Stralsund, Germany): *Annotation of Spatially Resolved Single-cell Data with STELLAR*.
- **Tan** and Cahan. International Society for Stem Cell Research Annual Meeting (2019, Los Angeles, CA): *SingleCellNet: a computational tool to classify single cell RNA-Seq data across platforms and across species*.

**Poster presentation**

- **Tan**, Hickey, Horowitz, and Nolan, Single Cell Genomics Gordon Conference (2022, Switzerland)
- **Tan** and Cahan. International Society for Stem Cell Research Annual Meeting (2020, Virtual)
- **Tan** and Cahan. International Society for Stem Cell Research Annual Meeting (2017, Boston)

**ACADEMIC SERVICES**

- 2026      **Session Chair SLAS 2027 International Conference and Exhibition**  
Invite speakers for Multi-omics Integration in Biomedical Discovery session
- 2025 -    **Young Research Representative for European Society for Spatial Biology (ESSB)**  
Engage young researchers in ESSB
- 2024 - 2026    **Co-chair for the Single-cell Genomics Gordon Research Seminar**  
Design research program and screen applications
- 2019 -    **Ad Hoc Reviewer:**  
Cell, Nature Communications, Nucleic Acid Research, Cell Report, Stem Cell Reports, Biostatistics, Annals of Applied Statistics, Communication biology, Molecular Systems Biology, PLOS Computational Biology, Frontiers in Genetics, Gene, Journal of Translational Medicine

**TEACHING & MENTORING****Classroom teaching**

- Instructor
- Nov 2022    **Single-cell Spatial-omics: with Applications to Stem Cell Engineering and Cancer Immunotherapy @Stanford**  
Develop a course for graduate students and postdocs for spatial biology
- Jan 2020    **Neuroscience: A single cell story @JHU**  
Developed a syllabus and taught for an undergraduate class on scRNA-seq analysis
- Jul 2019    **Biomedical Engineering: Topics in Stem Cell Biology @JHU**  
Developed a syllabus and taught stem cell biology, developmental biology and computational tools to 27 high school students
- Teaching assistant
- 2018        **Computational Stem Cell Biology @JHU**  
Assisted in developing and teaching of computational stem cell biology to 20 undergraduate and graduate students
- 2017-2019    **Computational Biology and Bioinformatics @JHU**  
Assisted 90 graduate students to master basic high throughput analysis tools
- 2018-2019    **Practical Genomics @JHU**  
Assisted in a large classroom setting to teach researchers how to apply genomics technique in research
- Guest lecture
- Nov 2024    **BIOINF 590 Image Processing and Advanced Machine Learning for Cancer Bioinformatics @ University of Michigan**

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 Teach graph neural networks and CODEX spatial analysis

Dec 2023 **BENG469 Single Cell Biology @Yale University**  
 Teach computational analysis and imaging-based spatial biology

2020 - 2022 **Computational Stem Cell Biology @JHU**  
 Teach machine learning algorithms on cell type classification

## PEDAGOGY AND LEADERSHIP TRAINING

|      |                                                                    |                                                                                                                                                                                                                                          |
|------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 | Arthur and Sandra Irving Cancer Immunology Symposium               | The symposium covers aspects on how excellent science can be advanced by formulating questions, designing and executing studies to address these questions, hiring, mentoring, collaborating, leadership, raising funds, and publishing. |
| 2024 | Stanford University Preparing for Faculty Careers Course           | The course explores different types of faculty roles and different institutional contexts and helps to develop a plan to thrive once you land a faculty position                                                                         |
| 2018 | Johns Hopkins University Preparing Future Faculty Teaching Academy | The program introduces evidence-based inclusive teaching practices, provide opportunities for mentored teaching & educational research,                                                                                                  |

## RESEARCH EXPERIENCES

|                    |                                                                                                                                                                                                                                                           |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feb-Jun2024        | <b>Visiting Scholar, EMBL, Germany with Dr. Anna Kreshuk</b><br>Devise a deep learning model for 3D segmentation of light microscopic images                                                                                                              |
| May-Aug2020        | <b>Summer internship, Merck Research Laboratory</b><br>Characterized spatial relationship between immune cells and cancer cells in renal cell carcinoma by developing and applying machine learning models                                                |
| Jun-Aug 2014       | <b>Vienna BioCenter Summer research with Dr. David Keays, Austria</b><br>Identified somatic mutations responsible for microcephaly from trio-exome sequencing and sequentially investigate expression of MAST1/2 during mouse embryonic brain development |
| Sept 2013-Jun 2014 | <b>Research assistant of Prof. Jiang Liwen, CUHK, Hong Kong</b><br>Characterized Rab7's roles in the protein trafficking (fusion of autophagosome to lysosome) via confocal imaging and sequential image analysis                                         |
| May-Aug 2013       | <b>Honorary fellow with Prof. Sebastian Bednarek, University of Wisconsin-Madison, USA</b><br>Identified and characterized protein-protein interaction (SCD1 and RabE) using genetic, biochemical, and imaging approaches                                 |
| Jan-May 2013       | <b>Research assistant of Prof. Carlos Aizenman, Brown University, USA</b><br>Characterized and statistically analyzed tectum neurons morphological changes after Valproic acid treatment with Confocal Microscopy and sequential image analysis           |

## COMMUNITY SERVICES

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|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022-now | <b>Students for Higher Education Opportunities and Representation in Training</b><br><u>Mentor</u><br>Provide mentorship to combatting socioeconomic and systemic inequalities in the medical and graduate school admissions processes |
| 2021     | <b>Cardiovascular Institute Summer School</b><br><u>Admission board member</u><br>Select candidates research internships from 460 applicants                                                                                           |

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- 2018 - 2019    **Biomedical Engineering Extramural Development in Graduate Education**  
    Internship Liaison  
    Matched interdisciplinary PhD students with desired internships  
    Hosted events for career development
- 2017 - 2019    **BCMB Colloquium Team**  
    Team Leader  
    Organized symposiums for PhD students (over 400 participants annually) to communicate their research  
    Designed survey and analyze data to give presenters quantitative feedback